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 Calculator

Lecture due Jan 24, 2024 20:30 IST

Key point of row echelon form : Matrices in row echelon form correspond to systems that are ready to be solved immediately by **back-substitution**. To perform back-substitution, you solve for each variable in reverse order (from bottom row to top row). (On the next page, we will see that you must introduce a parameter for each variable that can not be directly expressed in terms of later variables, and substitute values into earlier equations once they are known.)

Example problem The augmented matrix in row echelon form

$$\left(\begin{array}{ccc|c} 1 & -1 & 4 & 1 \\ 0 & 8 & -4 & 0 \\ 0 & 0 & 3 & 1 \end{array}\right)$$

describes a system of equations

$$\begin{aligned} x - y + 4z &= 1 \\ 8y - 4z &= 0 \\ 3z &= 1 \end{aligned}$$

Find the general solution to the system.

Solution: Solve the last equation first to get.

$$z = 1/3.$$

Substitute this into the second to last equation to get :

$$\begin{aligned} 8y - 4(1/3) &= 0 \\ y &= 1/6. \end{aligned}$$

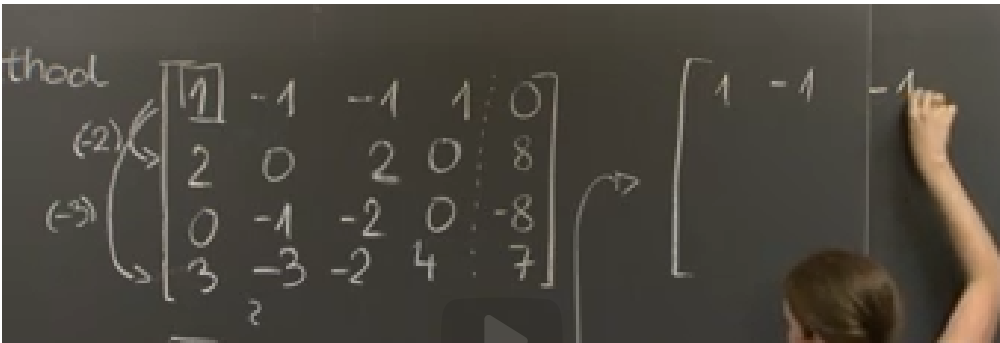
Now substitute both values into the first equation to get

$$\begin{aligned} x - 1/6 + 4/3 &= 1 \\ x &= 1 + 1/6 - 4/3 = -1/6. \end{aligned}$$

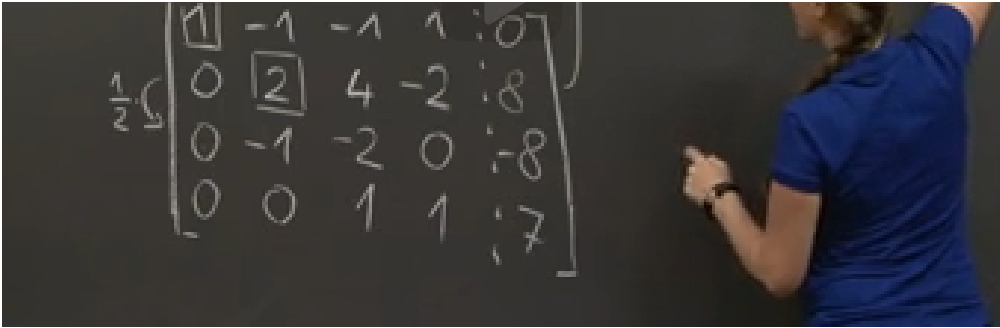
Conclusion: The general solution written as a column vector is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1/6 \\ 1/6 \\ 1/3 \end{pmatrix}.$$

Worked example: elimination and back substitution



Minus 3 minus 1 of--
sorry, 3 minus 2 is 1.
And minus 3 plus 4 is 1.
And there we go.
The first column looks like a first
column
of an upper triangular matrix.
Now, let's do the same to the
second column.



This is going to be our pivot, the number that we use to get rid of numbers under it, and we see that, to get rid of this number here, we will need to multiply it with 1/2. So multiply the whole second row with 1/2 and add it to the third row. The matrix that we get will have the first row the same



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Back-substitution practice

3/3 points (graded)
The augmented matrix in row echelon form

$$\left(\begin{array}{ccc|c} 2 & 3 & 5 & 7 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 3 & 9 \end{array}\right)$$

represents the system

$$\begin{pmatrix} 2 & 3 & 5 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \\ 9 \end{pmatrix}.$$

Use back-substitution to find the general solution.

$x =$

✓ Answer: 5

$y =$

✓ Answer: -6

$z =$

✓ Answer: 3

Solution:

The last equation says

$$3z = 9, \tag{2.4}$$

i.e., $z = 3$. Plugging this value into the second equation we get

$$y + 4(3) = 6, \tag{2.5}$$

which gives $y = -6$. Plugging the value of y and z into the first equation we get

$$2x + 3(-6) + 5(3) = 7, \tag{2.6}$$

which gives $x = 5$.

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You have used 1 of 4 attempts

